

Operacion Nexa 5G/6G - Anexo de calculos

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Escenario A - Distrito financiero

1. Ruido termico: $N = -93.990 \text{ dBm}$.
2. Sensibilidad: $\text{Sens} = -76.990 \text{ dBm}$.
3. Perdida maxima: $L_{\text{max}} = 125.990 \text{ dB}$.
4. Radio por cobertura: $R_{\text{cov}} = 0.553 \text{ km}$.
5. Trafico por usuario: $A_{\text{user}} = 0.100 \text{ Erl}$.
6. Densidad de trafico: $A_{\text{km}^2} = 250.000 \text{ Erl/km}^2$.
7. Recomendacion final: $N = 4$ con $R_{\text{design}} = 0.129 \text{ km}$.

Escenario B - Festival global

1. Ruido termico: $N = -93.990 \text{ dBm}$.
2. Sensibilidad: $\text{Sens} = -86.990 \text{ dBm}$.
3. Perdida maxima: $L_{\text{max}} = 135.990 \text{ dB}$.
4. Radio por cobertura: $R_{\text{cov}} = 3.412 \text{ km}$.
5. Radio por capacidad: $R_{\text{cap}} = 0.225 \text{ km}$.
6. Etapa de splitting recomendada: S4 con $R = 0.213 \text{ km}$.

Supuestos aplicados

| assumption | applied_value | justification |

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| Reference target area | 1 km² for both scenarios | The statement asks for the number of cells over a proposed target area but does not provide a fixed value. A common normalized 1 km² area is assumed to compare cell density without bias. |

| Channel partition in scenario A | floor(100 / N) channels per site, then uniform split across 3 sectors | The statement requests $N = 3, 4, 7$ and three sectors per site, but it does not define a more detailed PRB scheduler. The uniform split is the simplest reproducible assumption. |

| Interference approximation | 2 first-tier interferers for sectorized cells and 6 for omnidirectional reference | This is the standard first-tier hexagonal approximation used to compare reuse strategies in teaching-oriented planning exercises. |

| Hexagonal geometry | Regular hexagonal cell area model | Needed to convert traffic-limited area into an equivalent cell radius and then into the number of required sites. |

| Modern radio context | OFDMA/LTE/5G design logic with adaptive MCS | The statement explicitly asks for a modern mobile-network framing even though the numerical exercise is simplified to link budget and Erlang B. |